

TITANIC DATASET - DATA PREPARATION & CLEANING REPORT

Project Information

- **Project Title:** Titanic Dataset - Data Preparation & Cleaning
- **Role / Position:** Data Analyst Internship - Week 1 Task
- **Dataset Used:** Titanic-Dataset.xlsx (681 Records, 12 Features)
- **Final Deliverables:** Muhammad Waqar-Task-1-Data Analyst.ipynb & cleaned_titanic.csv

Executive Summary & Key Findings (10 Points)

1. **Dataset Inspection & Structure:** Initial dataset exploration revealed 681 entries and 12 features covering passenger demographic, ticket, and survival details.
2. **Data Quality Issues Identified:** Missing values were detected across three key features: Cabin (544 missing / ~80%), Age (147 missing / ~21.5%), and Embarked (1 missing). Zero duplicate rows were found in the dataset.
3. **Demographic Imputation (Age):** Instead of using a global mean, missing Age values were imputed using the group median based on Pclass and Sex to preserve realistic age distributions across passenger demographics.
4. **Categorical Imputation (Embarked & Cabin):** The single missing Embarked record was filled using the dataset mode ('S'). For Cabin, missing values were categorized as 'Unknown' while generating a new binary flag (Cabin_Known).
5. **Feature Engineering - Social Indicators:** Calculated Family_Size (SibSp + Parch + 1) and generated a binary Is_Alone indicator to analyze how traveling with family influenced survival chances.
6. **Feature Engineering - Title Extraction:** Extracted passenger honorifics/titles (e.g., *Mr*, *Mrs*, *Miss*, *Master*) from the Name column to support future predictive feature modeling.
7. **Column Standardization:** Transformed all dataframe column names into standard lowercase format (passengerid, survived, pclass, sex, etc.) to adhere to Python naming conventions.
8. **Gender Survival Dynamics:** Exploratory visual analysis highlighted a significant gender disparity: female passengers experienced a **77.18%** survival rate, compared to only **16.84%** for male passengers.
9. **Socio-Economic Survival Impact:** First-Class (Pclass 1) passengers had the highest survival rate at **56.46%**, followed by Second-Class at **44.29%**, and Third-Class at **23.86%**, reflecting socio-economic priority during rescue operations.
10. **Data Export & Readiness:** The final preprocessed data was exported to cleaned_titanic.csv, fully cleansed, missing-value-free, and ready for downstream analytical workflows.